

TECHNOLOGICAL UNIVERSITY (THANLYIN)
DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

**A PERSONALIZED RECOMMENDATION SYSTEM FOR
HOTEL BOOKING USING CONTENT-BASED FILTERING**

BY
MA THAZIN NYUNT
VI IT - 4

GRADUATION THESIS

OCTOBER, 2025
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A THESIS
SUBMITTED TO THE DEPARTMENT OF
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OCTOBER, 2025
THANLYIN

TECHNOLOGICAL UNIVERSITY (THANLYIN)
DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

We certify that we have examined, and recommend to the University Steering Committee for Bachelor of Engineering Graduation thesis entitled: **“A PERSONALIZED RECOMMENDATION SYSTEM FOR HOTEL BOOKING USING CONTENT-BASED FILTERING”** submitted by **MA THAZIN NYUNT**, Roll No. **VI IT - 4 (NOVEMBER, 2024)** to the Department of Information Technology Engineering as the requirement for the Bachelor of Engineering degree, **BE(IT)**.

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ABSTRACT

This study develops a personalized hotel booking system incorporating a content-based filtering recommendation engine to enhance user experience in online reservations. The system was developed using the MERN stack (MongoDB, Express.js, React, and Node.js) to ensure scalability, responsiveness, and usability for both customers and administrators. Core functionalities include secure user registration and authentication via Clerk, hotel and room listings with advanced filtering options, booking management, payment processing through Stripe, and automated email notifications using Nodemailer.

The main contribution of this work is the integration of a personalized recommendation system. For non-registered users, the platform displays top-rated hotels based on popularity and customer ratings to provide relevant suggestions without prior interactions. For registered users, the content-based filtering algorithm dynamically constructs user profiles by analyzing historical booking data, preferred amenities, price ranges, and ratings. Using similarity measures, the system ranks hotels and recommends those most closely aligned with each user's preferences. Additionally, an administrative dashboard supports hotel and booking management with role-based access for administrators and hotel owners.

Overall, the proposed system demonstrates that combining personalized recommendation techniques with secure and efficient booking functionalities can substantially enhance user satisfaction, decision-making, and operational efficiency in hotel reservation platforms.

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